1. Data Definitions V2

This section expands the M1 data definitions into full field-level definitions. The same names are used in the database, the API JSON, the TypeScript types (`lib/spatial/contracts.ts`), the UML in Section 5 and the UI. Each entity is tagged **[M2]** if it is built and used in the M2 prototype, **[M2 partial]** if only part of it is built, or **[Planned]** if it is built in M3 to M5. Planned entities are still fully defined here because they are part of the full design.

1.1 Conventions

Convention	Definition
Naming	`snake_case` in SQL and API JSON. Table names are plural (`graph_nodes`); entity names in the UML are singular (GraphNode).
Terms	A path segment is a walkable connection between two graph nodes. Section 2 uses the same term; the M1 requirements called it an *edge*.
IDs	UUID. Every reference points to an existing record.
Time	`timestamptz` (UTC), ISO 8601 in the API. Displayed in local time.
Units	Positions and distances in meters, latency in milliseconds, heading in degrees. Distances are stored to the millimeter and shown with one decimal (5.1 m).
Missing data	`null`, never 0. A missing latency, pose or error is never stored as zero.
Enums	Fixed value lists; unknown values are rejected by the database and the API. Exception: pose `data_origin` is not taken from the client as is. The server sets it to `synthetic` for the fake-test device or when the client sends `synthetic`, and to `real` otherwise (1.10).
Text limits	Names and labels 120 characters; descriptions and reasons 1,000 characters.
Map scope	Every spatial record belongs to one map version; linked records must share it and its graph revision.

Shared values

Value	Definition
`position`	`{x_m, y_m, z_m}` in world meters.
`orientation`	Quaternion `{x, y, z, w}`. Expected to be normalized; in M2 the API checks only that the four values are numbers, and normalization checks follow in M3.
`world_frame`	Object `{frame_id, origin_description, handedness: "right", up_axis: "y", units: "meters"}`. Right-handed, Y up, X and Z horizontal. Stored on each map version.
`provider_pose`	Raw phone pose as reported by the Niantic SDK in Unity: `{position: {x, y, z}, rotation: {x, y, z, w}}`.
`route_step`	`{segment_id, from_node_id, to_node_id}` (UUIDs); gives the travel direction explicitly.

How coordinates line up

Data	Frame	How it gets into the world frame
World frame (HSS 349)	VPS map frame of *SFSU HSS 349 v2*	This is the world frame of the HSS 349 map version.
Scanned 3D model (GLB)	Same VPS map frame	Exported from the VPS map, so no conversion is needed (identity).
Graph nodes, destinations	World frame	Recorded on site with the iPhone while localized on the same VPS map. `y_m = 0` is phone height; the viewer draws the graph at the floor height measured in the scan (−1.29 m).
Test points	World frame	The location is picked on the model. The reference position used to measure accuracy is measured independently on site, not taken from the model (1.8).

Data	Frame	How it gets into the world frame
iPhone pose from VPS	Unity coordinates (left-handed) of the same VPS map	Converted by the server: world = (x, y, −z). From M3, a map whose model and VPS frame differ uses a verified `frame_transforms` row (1.12).

1.2 Users, roles and privileges

[Planned] Application roles. The roles from M1 are kept. Accounts and passwords are managed by Supabase Auth and never stored in application tables. Permissions are enforced by the API (RoleGuard, Section 5) and by database row-level security, never by the UI alone.

Role	Can do
`visitor`	View the environment and destinations; navigate; see own session, pose and route.
`developer`	Visitor abilities, plus run localization attempts and labeled simulation or replay, and view site evaluation results.
`operator`	Visitor abilities, plus edit destinations and path segments, open or close segments, set robot access, and review agent reports.
`administrator`	Manage accounts, site memberships, site configuration and map publication.

[M2] Access in the prototype. M2 has no sign-in yet. Access is limited as follows:

Actor	Can do in M2
Anyone with the site URL	Through the API: read the published public demo maps (`sites.is_public_demo`), their graph, destinations and synthetic attempts, and request routes. Directly from Supabase (anon key): read pose samples. No writes. The current pose read policy allows reading every pose; limiting it to public demo maps is planned for M3.
iPhone app with the device token	Add pose samples through POST /api/poses. The token is a server-side environment variable (`DEVICE_TOKEN`), never stored in a table.
Team, TA and instructor (Supabase project members)	Full table access in the Supabase dashboard, for seeding and migrations.

user_profiles [Planned]

Field	Type	Notes
`user_id`	UUID	Same as the Supabase Auth user ID
`display_name`	text	Shown in the app, max 120 characters
`created_at`	timestamp	When the profile was created

site_memberships [Planned]

Field	Type	Notes
`membership_id`	UUID	Primary key
`user_id`	UUID	Links to `user_profiles`
`site_id`	UUID	Site the role applies to
`role`	enum	`visitor`, `developer`, `operator`, `administrator`; one active role per user per site
`is_active`	boolean	False = access removed
`created_at`	timestamp	When the membership was granted

1.3 Site [M2]

`sites`: a physical area. M2 has two published public demo sites:

- **SFSU HSS 349**: the real scanned classroom used for the 3D viewer, routing and live position.
- **SYNTHETIC M2 demo**: invented coordinates and labeled sample attempts for the list and search prototype (`is_test_site = true`). It is not a campus scan.

Field	Type	Notes
`site_id`	UUID	Primary key
`name`	text	e.g. "SFSU HSS 349"
`description`	text	Area covered
`is_test_site`	boolean	`true` for seeded test data
`is_public_demo`	boolean	`true` = readable without sign-in in M2
`active_map_version_id`	UUID, nullable	Published map used by default
`created_at`, `updated_at`	timestamp	Set by the database on insert and update

1.4 Map version [M2]

`map_versions`: one captured version of a site.

Field	Type	Notes
`map_version_id`	UUID	Primary key
`site_id`	UUID	Owning site
`version_label`	text	Unique per site, e.g. "HSS349-v1"
`publication_state`	enum	`draft`, `published`, `retired`; published requires `published_at`
`world_frame`	object	See 1.1
`visual_asset_id`	UUID, nullable	Displayed 3D model (1.12). Null in M2; the viewer loads the model file directly.
`vps_map_id`	text, nullable	Niantic localization map ID; separate from the visual asset
`vps_status`	enum	`not_configured`, `draft`, `production`, `unavailable`; draft and production require `vps_map_id`
`current_graph_revision`	integer	Topology version
`current_path_state_revision`	integer	Open/closed and robot-access version
`captured_at`, `published_at`	timestamp, nullable	When the area was scanned and when this version was published; published requires `published_at`

The visual model is what users see; the VPS map is what the phone localizes against. They are stored separately. For HSS 349 they share one frame (1.1); other maps are aligned through a frame transform from M3 (1.12).

1.5 Graph nodes [M2]

`graph_nodes`: waypoints that path segments and destinations connect to.

Field	Type	Notes
`node_id`	UUID	Primary key; referenced by path segments and destinations
`map_version_id`	UUID	Map version the node belongs to

Field	Type	Notes
`graph_revision`	integer	Graph revision the node belongs to; must match the map's current revision to be shown
`label`	text	e.g. "N1 Door"
`position`	position	World meters

HSS 349 has four nodes: N1 Door, N2 Back corner (door side), N3 Back corner (far) and N4 Front of room. The UI shows each node by the name of its destination (1.7).

1.6 Path segments [M2]

`path_segments`: walkable connections between two nodes.

Field	Type	Notes
`segment_id`	UUID	Primary key; referenced by route steps
`map_version_id`, `graph_revision`	UUID, integer	Same revision as both endpoints
`label`	text	Searchable, e.g. "Front to Door"
`start_node_id`, `end_node_id`	UUID	Different nodes
`direction`	enum	`bidirectional`, `start_to_end`
`distance_m`	number	Positive; must match the geometry length within 0.01 m
`geometry`	array of position	2 to 1,000 points; first and last point equal the start and end node positions
`availability`	enum	`open`, `closed`
`robot_access`	enum	`allowed`, `restricted`
`last_state_revision`	integer	Path-state revision of the last open/closed or robot-access change
`updated_at`	timestamp	Set by the database on every update

A human route may use any **open** segment. A robot route may use only segments that are **open and robot-allowed**. Reopening a segment keeps its robot access rule. The database rejects a segment whose geometry or distance does not match its nodes.

1.7 Destinations [M2]

`destinations`: named places users can navigate to. HSS 349 has four, one at each node: **Door, Back Corner (Door Side), Back Corner (Far Side)** and **Front of Room**. The synthetic demo map has three (Room 103, Lobby, East room).

Field	Type	Notes
`destination_id`	UUID	Primary key; sent in route requests
`site_id`, `map_version_id`	UUID	Site and map version the destination belongs to; must match its node
`graph_revision`	integer	Same revision as its node
`node_id`	UUID	Destination node in the same graph
`name`	text	Searchable; shown in the Start and Destination lists
`description`	text	Short explanation shown with the name, e.g. "Room entrance"
`is_active`	boolean	Offered for new navigation

Field	Type	Notes
updated_at	timestamp	Set by the database on every update

1.8 Test points [M2]

test_points: fixed physical camera positions used to evaluate localization. The full design has five evaluation points per site; points used for alignment are kept separate. In M2 the five test points belong to the synthetic demo map (is_synthetic = true) and are used only for the attempt search.

Field	Type	Notes
test_point_id	UUID	Primary key; used as a search filter
map_version_id	UUID	Map version the test point belongs to
label	text	Searchable, unique per map, e.g. "TP3 lobby"
purpose	enum	evaluation, alignment
reference_position	position, nullable	Camera position measured independently on site (not taken from the model)
reference_method	text, nullable	How it was measured
reference_uncertainty_m	number, nullable	Never assumed 0
reference_is_verified	boolean	Only verified points count toward error; requires a real point with position, method and uncertainty
is_synthetic	boolean	true for seeded test data

1.9 Localization attempts [M2]

localization_attempts: one record per started localization attempt, including every failure and timeout. **Success** means a valid pose for the intended map within 30 seconds; accuracy is measured separately.

Field	Type	Notes
attempt_id	UUID	Resending the same attempt keeps its ID; a new attempt gets a new ID
session_id	UUID, nullable	Links to a session from M3; null in M2
map_version_id, test_point_id	UUID	Map and test point of the attempt; the test point must belong to the map
device_id	text	Pseudonymous device label
sdk_version	text, nullable	Required for live attempts; null for synthetic data
mode	enum	live, simulation, replay
data_origin	enum	live_sdk, synthetic, recorded (pairs with mode: live/live_sdk, simulation/synthetic, replay/recorded)
condition_label	text	e.g. "daylight", "evening"
camera_direction	text	e.g. "facing north"
started_at, ended_at	timestamp	ended_at is null while in progress
outcome	enum	in_progress, success, timeout, tracking_loss, network_error, map_mismatch, cancelled, other_failure
latency_ms	number, nullable	Success only, 0 to 30,000
failure_reason	text, nullable	Required for failures

Field	Type	Notes
valid_pose_sample_id	UUID, nullable	Required for a live success from M3; null in M2
horizontal_error_m	number, nullable	X/Z distance to a verified reference; null in M2
error_is_valid	boolean	False for synthetic data and in M2
received_at	timestamp	Server receipt time

Horizontal error = √((pose.x − ref.x)² + (pose.z − ref.z)²). In M2 all attempts are labeled synthetic samples (`mode = simulation`), and the database itself keeps the M3 fields empty, so no sample can be mistaken for a measured result. Synthetic and replayed attempts never count as live evidence.

1.10 Pose samples [M2 partial] and sessions [Planned]

A **pose sample** is one position report from the phone. It is built in M2: the iPhone app sends one every 0.5 s and the /live page shows the latest one as a moving dot. A **session** groups one device's live, simulation or replay data from M3.

pose_samples [M2 partial]

Field	Type	Notes
sample_id	UUID	Created by the phone; sending the same ID again stores nothing new
map_version_id	UUID	Map the pose belongs to; must be published
attempt_id	UUID, nullable	Localization attempt the pose belongs to, if any
device_id	text	Pseudonymous device label, 1 to 120 characters; "fake-test" for pipeline tests
provider_map_id	text, nullable	VPS map the phone localized against; must match `vps_map_id` when both are set
provider_pose	provider_pose	Raw pose from the phone (1.1)
world_position	position, nullable	Converted by the server: (x, y, −z)
heading_deg	number, nullable	0 to 360; 0 = world +Z, increasing toward world +X
tracking_status	enum	`localizing`, `valid`, `lost`
data_origin	enum	`real`, `synthetic`; decided by the server: synthetic for the fake-test device or when the client sends synthetic, otherwise real
captured_at	timestamp	Time on the phone; at most 1 minute in the future
received_at	timestamp	Time on the server
session_id, transform_id, world_orientation	UUID, UUID, orientation	Planned for M3

A pose is **live** when its tracking is `valid` and it is less than 5 seconds old; otherwise /live shows it as “Not live”. Synthetic poses are labeled “test pose” on /live but, in M2, still drive the live dot and route. From M3, stale, lost and synthetic poses cannot start a new live route.

navigation_sessions [Planned]

Field	Type	Notes
session_id	UUID	Primary key
user_id	UUID, nullable	Null only for seeded test sessions
site_id, map_version_id	UUID	Map is fixed for the whole session

Field	Type	Notes
device_id	text	Pseudonymous device label
device_model	text	e.g. "iPhone 17 Pro"; "synthetic" for test data
sdk_version	text, nullable	Niantic SDK version used
mode	enum	live, simulation, replay
data_origin	enum	live_sdk, synthetic, recorded
status	enum	active, ended, failed
started_at	timestamp	When the session started
ended_at	timestamp, nullable	Null while active

1.11 Routes [M2 partial] and simulated agent [Planned]

In M2, POST /api/route computes a route from a start node to a destination, for a person or the simulated robot, and returns it without storing it. **From M3**, routes are stored in routes so they can be invalidated and replanned.

Field	Type	Notes
route_id	UUID	M3+ (stored routes only)
session_id	UUID, nullable	M3+
map_version_id	UUID	Map the route was computed on
graph_revision, path_state_revision	integer	Graph and open/closed state used
start_mode	enum	manual in M2; live from M3
start_node_id	UUID	M2: chosen by hand on /viewer, or the node nearest the phone on /live
start_position	position, nullable	M3+: live position used as the start
start_segment_id	UUID, nullable	M3+: walkable segment the live start is attached to
destination_id	UUID	Destination requested; resolved to its node by the API
profile	enum	human, robot
steps	array of route_step	Ordered path; empty when unreachable or when start = destination
total_distance_m	number, nullable	0 when start = destination; null if unreachable
status	enum	valid, unreachable (M2); invalidated, arrived (M3+)
created_at	timestamp	When the route was computed (server time)

simulated_agents [Planned]

Field	Type	Notes
agent_id	UUID	Primary key
session_id	UUID	Simulation-mode session only
route_id	UUID, nullable	Current route
position	position	Current position on its path
current_step_index	integer, nullable	Index into the route's steps
step_progress_m	number	Distance along the current step
speed_mps	number	Configured simulation speed (> 0)

Field	Type	Notes
`state`	enum	`idle`, `running`, `paused`, `waiting_for_operator`, `arrived`, `unreachable`
`updated_at`	timestamp	When the position or state last changed

A live start (M3) is accepted only if it is within `max_start_distance_m` (a site setting, agreed after first tests) of a walkable segment on the same map version and the pose is live; otherwise it is rejected with an error. The robot is always labeled SIMULATED in the UI. When a segment's availability or robot access changes, affected routes are invalidated and replanned, or marked unreachable.

1.12 Visual assets [M2 partial] and alignment [Planned]

In M2, the HSS 349 model is a static GLB file served by Vercel (`public/models/hss349-v2.glb`, 4.9 MB, textures 2048 px WebP, reduced from the 14.5 MB export). No frame transform is needed because the model, the graph and the poses share the VPS frame (1.1). Both tables below are created in M3, when models move to Supabase Storage and more maps are added.

visual_assets [Planned]

Field	Type	Notes
`asset_id`	UUID	Primary key
`map_version_id`	UUID	Map version the asset displays
`asset_kind`	enum	`mesh`, `preview_image`
`storage_path`	text	Supabase Storage key
`file_format`	enum	`glb`, `png`, `jpeg`, `webp`
`byte_size`	integer	Must be within the limits below
`checksum_sha256`	text	64 hex characters
`license_reference`	text	Who captured it / permission source
`created_at`	timestamp	When the asset was uploaded

frame_transforms [Planned]

Field	Type	Notes
`transform_id`	UUID	Primary key
`map_version_id`	UUID	Map version the transform belongs to
`source_frame_id`	text	e.g. a VPS site frame
`target_frame_id`	text	The map's world frame
`matrix_4x4`	array of 16 numbers	Row-major source → world transform; must be invertible
`validation_status`	enum	`unverified`, `verified`, `rejected`
`verified_at`	timestamp, nullable	When the transform was checked; required when verified
`verification_notes`	text	Which known points were checked and the error found

Media formats and limits

Asset	Format	Max size
3D mesh	GLB (textures JPEG or WebP, max 4096 px)	100 MB per file; keep under 10 MB for phones
Images	PNG, JPEG, WebP	10 MB
Video	Not stored	n/a

Total spatial asset storage stays within the 1 GB budget from M1. Raw camera recordings are not stored. From M3, live navigation uses only `verified` transforms.

1.13 Path events, simulated obstacles and agent reports [Planned]

path_events

Field	Type	Notes
`event_id`	UUID	Primary key
`map_version_id`, `segment_id`	UUID	Map version and segment that changed
`actor_user_id`	UUID	Operator or admin who made the change
`event_type`	enum	`availability_change`, `robot_access_change`
`old_value`, `new_value`	text	Must be valid values of that field and must differ
`reason`	text	Why the change was made, e.g. "wet floor"; required
`path_state_revision`	integer	New map-wide revision number
`source_report_id`	UUID, nullable	Agent report that caused it, if any
`committed_at`	timestamp	When the change was committed

agent_reports

Field	Type	Notes
`report_id`	UUID	Primary key
`agent_id`, `segment_id`	UUID	Robot that reported it and the blocked segment
`reason`	text	Tester-injected simulated obstacle, not real sensing
`status`	enum	`pending`, `confirmed`, `rejected`
`reported_at`	timestamp	When the robot sent the report
`reviewed_by_user_id`	UUID, nullable	Required once reviewed
`reviewed_at`	timestamp, nullable	When the operator decided; required once reviewed
`closure_event_id`	UUID, nullable	Required when confirmed

simulated_obstacles

Field	Type	Notes
`obstacle_id`	UUID	Primary key
`session_id`	UUID	Simulation session
`segment_id`	UUID	Segment the tester blocked
`placed_by_user_id`	UUID	Tester who placed it
`is_active`	boolean	Set false when dismissed
`placed_at`	timestamp	When the tester placed it
`dismissed_at`	timestamp, nullable	When it was dismissed; null while active

When the robot reaches the node before a segment with an active obstacle, it stops (`waiting_for_operator`) and creates an agent report. **Confirm:** the segment closes and routes replan. **Reject:** the obstacle is dismissed (`is_active = false`) and the robot resumes if its route is still valid. A pending report never closes a path. Confirmation creates exactly one closure event; confirming twice creates no duplicate. Stale writes are rejected.

1.14 Metric snapshot and localization health [Planned]

metric_snapshots

Field	Type	Notes
`snapshot_id`	UUID	Primary key
`test_point_id`, `map_version_id`	UUID	Health is per point, device and map
`device_id`	text	Device the health applies to
`attempt_ids`	array of UUID	Exact attempts used
`attempt_count`	integer	All completed live attempts, including failures; in_progress excluded
`success_count`	integer	Attempts with outcome success
`success_rate`	number, nullable	0–1; null if no attempts
`median_latency_ms`	number, nullable	Successful attempts only
`valid_error_count`	integer	Attempts with a valid reference error
`median_horizontal_error_m`	number, nullable	Null if no valid errors
`health_status`	enum	`gray`, `red`, `yellow`, `green`
`calculated_at`	timestamp	When the snapshot was computed

Health uses only **completed live** attempts (in_progress is excluded) from the same test point, device and map version. The rules are applied in order and the first match wins:

Order	Status	Rule
1	Gray	Fewer than 10 attempts
2	Red	≥10 attempts and success below 60%
3	Red	≥10 attempts and ≥5 valid error samples and median error above 1.5 m
4	Gray	≥10 attempts but fewer than 5 valid error samples
5	Green	≥10 attempts, ≥5 valid error samples, success ≥80% and median error ≤1 m
6	Yellow	All other cases

Counts are always shown with the status. Health describes tested positions only, not map freshness.

1.15 AI summaries and exports [Planned]

ai_summaries

Field	Type	Notes
`summary_id`	UUID	Primary key
`snapshot_id`	UUID	Metrics the summary explains
`requested_by_user_id`	UUID	User who asked for the summary
`model_version`, `prompt_version`	text	Gemini model and prompt template used, for reproducibility
`status`	enum	`pending`, `completed`, `failed`, `timed_out`
`observations`	array of {text, metric_fields, attempt_ids}	Each claim cites its source
`suggestions`	array of text	Labeled separately from observations

Field	Type	Notes
`requested_at`	timestamp	When the summary was requested
`completed_at`	timestamp, nullable	When Gemini answered or the request failed; null while pending

attempt_exports

Field	Type	Notes
`export_id`	UUID	Primary key
`requested_by_user_id`	UUID	User who requested the export
`filters`	object	Test point, outcome, mode and date range used
`file_format`	enum	`csv`, `json`
`row_count`	integer	Number of attempts in the file
`generated_at`	timestamp	When the file was generated

Gemini receives only aggregate metrics and attempt IDs, never camera data. All numbers are computed by the server; generated numbers that do not match the snapshot are rejected. The dashboard keeps working if Gemini fails (15-second timeout).

1.16 Relationships

- A **site** has map versions; a **map version** owns its graph nodes, path segments, destinations, test points, pose samples, assets and transforms.
- A **path segment** connects two graph nodes in the same graph revision; a **destination** points to one node.
- A **localization attempt** belongs to one test point and map version (and to a session from M3).
- A **pose sample** belongs to one map version and optionally to one attempt (and to a session from M3).
- A **route** uses ordered path segments from one graph revision and one path-state revision.
- An **agent report** targets one path segment; confirmation links it to one **path event**.
- A **metric snapshot** records the exact attempts it was computed from; an **AI summary** cites one snapshot.
- History is preserved: attempts, events and referenced map versions are retired, not deleted.

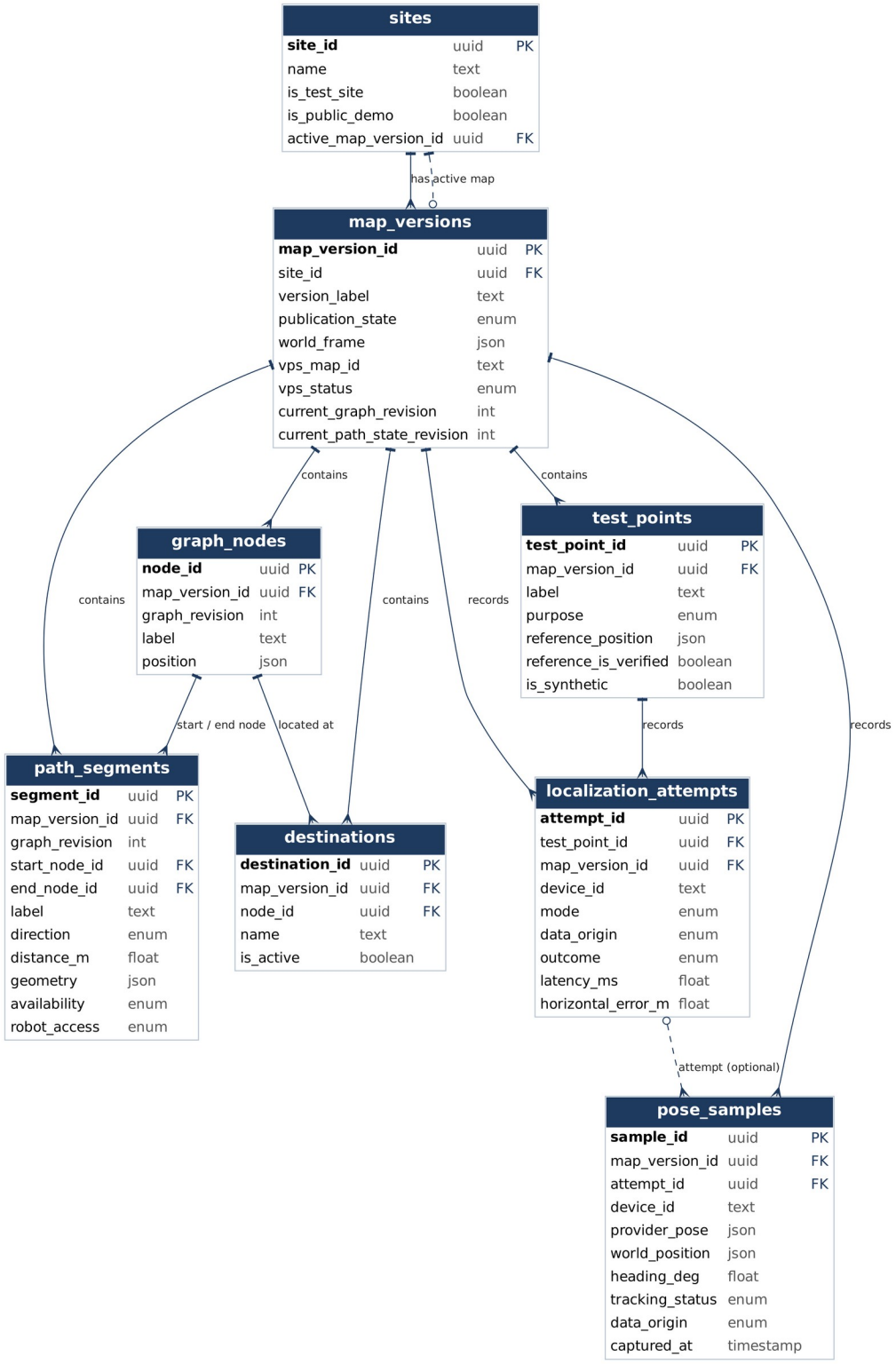


Figure 1.1: M2 database tables (key fields)

1.17 Traceability to requirements

Data concept	Section	Requirements (Section 2)
Users, roles and privileges	1.2	R1.1–R1.4, R7.1–R7.3, R8.1–R8.2
Site and map version	1.3–1.4	R2.1, R2.2
Navigation graph	1.5–1.6	R2.3, R5.2, R5.4, R7.1, R7.2, R8.1–R8.3, R15.3, R17.1, R17.2
Destination	1.7	R2.4, R5.1, R7.3, R15.3
Test point and reference	1.8	R12.1, R14.3
Localization attempt	1.9	R3.4, R12.1, R12.2, R14.1–R14.4, R15.1
Pose sample and session	1.10	R3.2, R3.3, R3.5, R4.1, R4.2, R6.2–R6.5, R11.1, R11.2
Route and simulated agent	1.11	R5.1–R5.5, R6.1, R9.1–R9.3, R10.1–R10.5
Visual asset and transform	1.12	R2.2, R3.3
Path event, obstacle and report	1.13	R8.1–R8.3, R17.2, R18.1–R18.6
Metric snapshot and health	1.14	R13.1, R13.2, R14.1–R14.5
AI summary and export	1.15	R15.2, R16.1